NATURAL LANGUAGE PROCESSING (NLP)

Project Milestone Handbook

Student Guide

Research Track & Industry Track

Table of Contents

1. Introduction

2. Choosing Your Project Track

3. Milestone Roadmap

4. Detailed Milestones

    Research Track

    Industry Track

5. Final Submission Checklist

    Research Track Checklist

    Industry Track Checklist

1. Introduction

Welcome to the NLP Project Milestone Handbook. This guide explains what you need to submit at each stage of your semester-long NLP project.

Before you begin, you must choose ONE project pathway:

- Research-Based NLP Project
- Industry-Based NLP Project

Both pathways follow six milestone checkpoints, and your progress is reviewed continuously throughout the semester so you receive feedback while there is still time to act on it. The sections that follow describe each track, the milestone roadmap, and exactly what to submit at every checkpoint.

2. Choosing Your Project Track

Each track offers a different kind of project experience. Read both descriptions carefully and choose the one that best matches your interests and goals.

Research Track	Industry Track
• Paper-driven investigation • Reproduce an existing paper • Identify a research gap • Create your own hypothesis • Perform controlled experiments • Write an IEEE-style paper	• Solve a real-world business problem • Build an end-to-end NLP solution • Use public or licensed datasets • Develop and evaluate a deployable system • Demonstrate business impact

3. Milestone Roadmap

Both tracks follow the same six-milestone structure, shown below.

Milestone	Research Track	Industry Track
M1	Literature Survey	Problem/ Idea Definition
M2	Baseline Reproduction	Data preprocessing
M3	Research Gap & Hypothesis	Model Development
M4	Controlled Experiments	Evaluation & Optimization
M5	Failure Analysis	Deployment Prototype
M6	Final Paper & Defence	Final Demo & Report (Preferred PPT)

4. Detailed Milestones

Research Track

The Research Track guides you through a paper-anchored investigation, from literature survey to final defence.

M1 — Literature Survey & Anchor Paper

Objective

Build a solid understanding of your chosen research area by surveying existing work and selecting one anchor paper that your project will reproduce and extend.

Student Deliverables

- Five related papers
- Structured comparison
- Anchor paper selection
- Reproducibility checklist

Expected Outcome

A clearly justified anchor paper and a structured comparison of related work that set the foundation for your reproduction in M2.

M2 — Baseline Reproduction

Objective

Reproduce the anchor paper's baseline so you gain hands-on familiarity with its dataset, method and results before proposing any extension.

Student Deliverables

- Dataset documentation
- Preprocessing pipeline
- Baseline implementation
- Original vs reproduced results
- Reproduction observations

Expected Outcome

A working, documented reproduction of the baseline, with an honest comparison between your results and the original paper's results.

M3 — Research Gap & Hypothesis

Objective

Identify a genuine gap in the anchor paper or related literature, and propose one clear, testable hypothesis to address it.

Student Deliverables

- Research gap
- One hypothesis
- Prediction before experiment
- Experimental plan

Expected Outcome

A well-defined hypothesis and experimental plan, with your prediction recorded before you begin experimenting.

M4 — Controlled Experiments

Objective

Test your hypothesis through controlled experiments that isolate the effect of the variable you are investigating.

Student Deliverables

- Baseline configuration
- Experimental configuration
- Metrics
- Predicted vs actual comparison
- Ablation study

Expected Outcome

Quantitative results comparing your baseline and experimental configurations, plus an ablation study isolating individual components.

M5 — Failure Analysis

Objective

Examine where and why your system fails, and reflect honestly on the limitations of your approach.

Student Deliverables

- Representative failure cases
- Evidence-based explanations
- Limitations
- Threats to validity

Expected Outcome

An evidence-based account of your system's failure modes, together with a clear discussion of limitations and threats to validity.

M6 — Final Paper & Defence

Objective

Consolidate your work into a complete IEEE-style paper and defend your findings orally.

Student Deliverables

- IEEE paper
- Reproduced vs original contribution
- References
- Presentation
- Viva

Expected Outcome

A polished paper that clearly separates reproduced work from your own contribution, presented and defended with confidence.

Industry Track

The Industry Track guides you through building and delivering a real-world, end-to-end NLP solution.

M1 — Problem Definition & Business Case

Objective

Frame a real business problem as a well-scoped NLP task, and define how success will be measured before any building begins.

Student Deliverables

- Problem brief
- User identification
- Business objective
- Success metrics
- Feasibility analysis

Expected Outcome

A concise problem brief with clearly identified users, a stated business objective and measurable success metrics that will guide the rest of the project.

M2 — Data Understanding & Preparation

Objective

Understand, clean and prepare your data responsibly so it is ready for modelling.

Student Deliverables

- Data card
- Exploratory analysis
- Data cleaning
- Statistical Analysis
- Privacy considerations

Expected Outcome

A documented, leak-free dataset with clean splits, ready for baseline modelling, and with privacy considerations addressed.

M3 — Baseline & Model Development

Objective

Establish a fair baseline and iterate toward a stronger modelling approach.

Student Deliverables

- Baseline model
- Candidate approaches
- Version-controlled implementation
- Experiment log

Expected Outcome

A working baseline plus at least one improved candidate approach, tracked in a version-controlled repository with an experiment log.

M4 — Evaluation & Optimization

Objective

Evaluate your model rigorously, understand its errors in depth, and improve it responsibly.

Student Deliverables

- Model Optimization
- Model Inferencing
- Improvement iterations
- Robustness testing

Expected Outcome

A documented evaluation with categorised errors, a record of improvement iterations, and evidence of robustness testing.

M5 — Deployment Prototype

Objective

Turn your model into a working, monitored system that could plausibly run in production.

Student Deliverables

- Working application/API
- Model deployment.
- Monitoring strategy
- Failure handling

Expected Outcome

A deployed, functioning prototype with measured performance and a clear plan for monitoring it and handling failures.

M6 — Final Delivery

Objective

Deliver your completed project as a professional, business-ready package.

Student Deliverables

- Technical report
- GitHub repository
- Live demonstration
- Business presentation
- Viva

Expected Outcome

A complete handover package — report, code and live demonstration — presented persuasively to a business audience.

5. Final Submission Checklist

Research Track Checklist

M1 — Literature Survey

- ☐ Five papers identified
- ☐ Structured comparison
- ☐ Anchor paper selection
- ☐ Reproducibility checklist

M2 — Baseline Reproduction

- ☐ Dataset documentation
- ☐ Preprocessing pipeline
- ☐ Baseline implementation
- ☐ Original vs reproduced results
- ☐ Reproduction observations

M3 — Research Gap & Hypothesis

- ☐ Research gap
- ☐ One hypothesis
- ☐ Prediction recorded before experiment
- ☐ Experimental plan

M4 — Controlled Experiments

- ☐ Baseline configuration
- ☐ Experimental configuration
- ☐ Metrics
- ☐ Predicted vs actual comparison
- ☐ Ablation study

M5 — Failure Analysis

- ☐ Representative failure cases
- ☐ Evidence-based explanations
- ☐ Limitations
- ☐ Threats to validity

M6 — Final Paper & Defence

- ☐ IEEE paper
- ☐ Reproduced vs original contribution
- ☐ References
- ☐ Presentation
- ☐ Viva

Industry Track Checklist

M1 — Problem Definition & Business Case

- ☐ Problem brief
- ☐ User identification
- ☐ Business objective
- ☐ Success metrics
- ☐ Feasibility analysis

M2 — Data Understanding & Preparation

- ☐ Data card
- ☐ Exploratory analysis
- ☐ Data cleaning
- ☐ Train / validation / test split
- ☐ Privacy considerations

M3 — Baseline & Model Development

- ☐ Baseline model
- ☐ Candidate approaches
- ☐ Version-controlled implementation
- ☐ Experiment log

M4 — Evaluation & Optimization

- ☐ Evaluation report
- ☐ Error analysis
- ☐ Improvement iterations
- ☐ Robustness testing

M5 — Deployment Prototype

- ☐ Working application / API
- ☐ Performance measurements
- ☐ Monitoring strategy
- ☐ Failure handling

M6 — Final Delivery

- ☐ Technical report
- ☐ GitHub repository
- ☐ Live demonstration
- ☐ Business presentation
- ☐ Viva